

TAICHUNG INTL, Taiwan

WMO: 467700

Lat: 24.265N

Lon: 120.621E

Elev: 663

StdP: 14.35

Time Zone: 8.00 (E08)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.6	(c) 48.5	(d) 37.3	(e) 33.5	(f) 49.2	(g) 41.0	(h) 38.8	(i) 51.3	(j) 31.0	(k) 48.6	(l) 27.7	(m) 51.3	(n) 16.2	(o) 20	(p) 0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	91.1	80.0	89.8	79.6	89.2	79.5	82.1	87.6	81.1	87.1	80.4	86.6	10.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	163.5	86.1	79.2	155.4	85.0	78.9	153.7	84.8	46.7	87.5	45.5	87.5	44.8	86.9	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.3	(b) 23.5	(c) 20.9	(e) 41.8	(f) 94.1	(g) 5.8	(h) 2.2	(i) 37.6	(j) 95.6	(k) 34.2	(l) 96.9	(m) 30.9	(n) 98.1	(o) 26.7	(p) 99.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.6	60.7	60.8	64.7	71.7	77.0	80.9	82.8	81.9	80.0	75.4	70.4	64.0
	DBStd	9.12	5.05	6.00	6.23	5.61	4.29	2.95	2.01	2.28	3.13	3.76	4.61	5.00
	HDD50	5	3	1	0	0	0	0	0	0	0	0	0	1
	HDD65	481	148	144	86	14	1	0	0	0	0	0	11	77
	CDD50	8246	335	303	455	651	836	926	1018	989	900	787	611	435
	CDD65	3245	15	25	75	215	372	476	553	524	450	322	172	46
	CDH74	27028	28	77	290	1091	2785	4621	6021	5387	3996	1965	660	107
	CDH80	8405	1	7	34	197	729	1529	2257	1823	1242	478	100	8
Wind	WSAvg	9.3	11.7	11.0	9.4	8.1	6.9	8.1	7.9	7.4	7.9	10.7	10.9	12.1
Precipitation	PrecAvg	59.3	1.3	3.3	3.8	5.7	7.7	10.6	8.6	9.4	5.7	1.2	1.0	1.2
	PrecMax	88.4	4.2	13.3	8.1	20.4	18.9	26.3	22.9	23.4	28.4	9.1	6.6	3.3
	PrecMin	27.4	0.1	0.1	1.2	0.2	0.9	0.9	1.4	1.0	0.5	0.0	0.0	0.0
	PrecStd	16.5	0.9	3.2	2.1	4.5	4.9	6.3	5.9	6.4	7.3	2.0	1.5	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	80.5	83.8	87.4	89.8	91.3	92.9	91.6	91.5	89.9	85.9	80.8
		MCWB	66.5	69.9	73.1	77.6	78.7	79.9	81.0	80.9	79.7	77.7	74.7	69.0
	2%	DB	73.6	76.1	80.3	84.3	87.8	89.7	91.2	89.9	89.6	87.4	82.6	77.0
		MCWB	64.9	68.1	71.4	76.1	78.6	79.5	80.0	80.3	78.9	77.8	73.3	68.7
	5%	DB	70.6	73.2	77.3	82.4	86.0	88.3	89.8	89.3	87.9	84.4	80.6	73.7
		MCWB	63.6	66.4	69.4	74.9	78.2	79.4	79.6	80.1	79.0	75.6	71.6	66.3
	10%	DB	68.1	69.9	73.8	80.2	84.2	87.6	89.1	87.8	86.2	82.5	78.5	71.6
		MCWB	62.0	64.1	67.8	73.6	77.2	79.3	79.6	80.1	78.6	74.2	70.4	65.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	72.6	74.9	78.9	81.0	82.4	83.7	83.9	82.2	80.3	76.7	72.0
		MCDB	73.8	77.4	81.4	84.1	86.4	86.9	88.8	88.9	87.5	86.2	83.3	78.0
	2%	WB	66.7	69.4	72.7	77.2	79.6	81.1	82.0	82.0	80.7	78.7	74.8	69.5
		MCDB	71.9	74.2	78.2	82.5	85.4	86.8	88.1	87.7	86.6	84.4	81.0	74.9
	5%	WB	65.0	67.0	70.8	75.8	78.6	80.3	80.9	80.9	79.8	77.7	73.2	67.7
		MCDB	69.6	71.5	75.8	80.7	84.1	86.4	87.5	86.8	85.5	83.0	78.5	72.5
	10%	WB	63.3	65.0	68.9	74.5	77.7	79.5	80.0	80.1	78.9	76.3	71.6	66.1
		MCDB	67.1	69.5	73.0	79.1	83.0	85.4	86.5	85.7	84.3	81.1	76.1	70.4
Mean Daily Temperature Range	5% DB	MDBR	13.5	13.8	13.7	12.8	11.8	11.4	12.3	11.4	12.3	13.9	14.3	14.0
		MCDBR	16.3	18.5	18.0	14.7	12.9	12.8	13.4	13.1	13.3	15.0	16.6	16.6
		MCWBR	9.9	11.5	10.6	9.0	7.3	6.6	6.6	6.5	6.7	8.2	9.0	10.2
	5% WB	MCDBR	14.8	18.1	17.2	13.5	12.0	12.1	12.9	12.2	12.3	13.5	15.4	16.1
		MCWBR	9.8	12.3	11.1	9.0	7.7	7.3	7.2	7.1	7.2	8.4	9.4	10.6
Clear-Sky Solar Irradiance	taub		0.486	0.540	0.656	0.666	0.570	0.524	0.537	0.568	0.558	0.522	0.493	0.486
	taud		2.017	1.902	1.667	1.699	1.969	2.118	2.073	1.976	1.977	2.021	2.081	2.034
	Ebn at Noon		240	236	216	218	239	248	245	238	237	238	237	234
	Edh at Noon		49	59	79	78	59	50	53	58	57	52	46	47
All-Sky Solar Radiation	RadAvg		911	1004	1166	1287	1484	1637	1869	1694	1553	1367	1020	903
	RadStd		128	148	129	174	226	178	115	164	148	117	98	110

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.45	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	+229	+202

Nomenclature: See separate page